

Malek Ahmad | majobs112002@gmail.com | +972549498961 | [Linkedin](#) | [Github](#) | [Portfolio](#)

Software Engineer proficient in **C/C++ and Python**, with strong experience in **system-level programming, parallel algorithms, and Linux-based environments**. Skilled in designing, developing, and optimizing high-performance software and AI-driven solutions. A **fast and self-driven learner** with strong **problem-solving skills, highly motivated team player**, and able to **work effectively under pressure**.

Skills

Programming languages: C, C++, Python, JavaScript.

IDEs: CLion, PyCharm, VS Code, Cursor.

Operating Systems: Linux (processes, memory, system calls), Windows.

Development Tools: Git, pytest, Valgrind.

Database: SQL (MySQL), Firestore, MongoDB, SQLite.

Education

2021 –2025: Azrieli College of engineering | Software engineering | GPA 90.83

Courses: C/C++ Programming | Operating Systems | Data Structures | Algorithms | Object Oriented Programming | Parallel and distributed computation | Reinforcement Learning

Projects

- **2025: Retrieval-Augmented Generation (RAG) System for Multilingual Document Search**, [Code link](#)

Built a scalable **RAG web application** for internal use in academic institutions, enabling **accurate Q&A** over large, **interconnected document collections** in **Hebrew, Arabic, and English**.

Developed the backend using **Python (Flask)**, **LangChain**, and **FAISS**. Designed a **React-based frontend** with a chat interface and file upload system.

- **2025: Reinforcement Learning AI for SOS Game**, [Code link](#)

Developed a **self-learning AI agent** for the SOS game using **Monte Carlo Tree Search (MCTS)** and **Puct** algorithms. Built using **Python, TensorFlow/Keras**.

- **2025: Parallel Image Matching System**, [Code link](#)

Implemented a **hybrid parallel image-matching system** in **C** using **MPI** for distributed multi-process execution and **OpenMP** for intra-process thread parallelism, with **dynamic workload** distribution across cluster nodes.

- **2024: Trainers' app for the Jerusalem Chess Center**, [Code link](#)

Developed a **React Native app** as part of a **4-person team**, connecting to a **Firestore database** to manage teachers, student groups, schools, and lessons.

Implemented backend logic for **CRUD operations, data synchronization, and real-time updates**.

Collaborated on **feature planning, architecture, and testing** to ensure reliable and scalable functionality.

Certificates

- **2025: Place-IL Developers Internship Program**

- Successfully passed a **competitive** selection process (**250+ candidates**), including **coding challenges** in **OOP, data structures, and databases**.

- Successfully completed **HR, technical, and group** interview stages.

- **2024: CS50: Introduction to Artificial Intelligence with Python – Harvard University (edX)**

- Completed an intensive course covering **search algorithms, optimization techniques, knowledge representation, and reinforcement learning**.

- Implemented projects involving **algorithmic problem solving, constraint satisfaction, and performance-aware solutions** in **Python**.

- Certificate and projects available on [GitHub](#).

Experience

2022 – 2025: Tutor | Azrieli College of Engineering | Jerusalem

Tutored **4-5 students per year** in **programming, algorithms, math** and **physics** courses, improving my responsibility, self-confidence, and socialization skills.

Languages

Arabic (Native) | Hebrew (Fluent) | English (High level)